CapstoneDesign-SupplyChain Project

BUSINESS REPORT

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Problem Statement

Business Context:

A fast-moving consumer goods (FMCG) company that entered the instant noodles market two years ago is facing challenges in balancing demand and supply across its warehouses nationwide. In some regions, demand far exceeds supply, while in others, excess inventory is leading to wastage and higher costs. This mismatch results in significant inventory losses and missed sales opportunities. To remain competitive and profitable in the highly dynamic FMCG sector, the company needs a data-driven approach to optimize its supply chain and align production with regional demand patterns.

Objective

As a data scientist, you are tasked with building a predictive model using historical sales and supply data to determine the optimum quantity of product to ship to each warehouse. Additionally, analyzing demand patterns across different regions will help management design targeted advertisement campaigns. Solving this problem will enable the company to minimize inventory costs, reduce stockouts, and drive higher sales through better demand forecasting and efficient supply chain management.

Data Description:

The dataset contains information about the historical sales patterns. The detailed data dictionary is given below:

- **Ware_house_ID**: Product warehouse ID
- **WH_Manager_ID**: Employee ID of warehouse manager
- **Location_type**: Location of the warehouse, like in a city or a village
- **WH_capacity_size**: Storage capacity size of the warehouse
- **zone**: Zone of the warehouse
- **WH_regional_zone**: Regional zone of the warehouse under each zone
- **num_refill_req_l3m**: Number of times refilling has been done in the last 3 months
- **transport_issue_l1y**: Any transport issue, like an accident or goods stolen, reported in the last year
- **Competitor_in_mkt**: Number of instant noodles competitors in the market
- **retail_shop_num**: Number of retail shops that sell the product under the warehouse area
- **wh_owner_type**: Company owns the warehouse, or they have rented the warehouse
- **distributor_num**: Number of distributors working between the warehouse and retail shops
- **flood_impacted**: Warehouse is in the flood-impacted area indicator
- **flood_proof**: Warehouse is flood-proof. Like storage is at some height, not directly on the ground
- **electric_supply**: The Warehouse has an electric backup, like a generator, so they can run the warehouse in load shedding
- **dist_from_hub**: Distance between the warehouse to the production hub in Kms
- **workers_num**: Number of workers working in the warehouse
- **wh_est_year**: Warehouse established year
- **storage_issue_reported_l3m**: Warehouse reported a storage issue to the corporate office in the last 3 months. Like rats, fungus due to moisture, etc.

Data Description

- **temp_reg_mach**: Warehouse has a temperature regulating machine indicator
- **approved_wh_govt_certificate**: What kind of standard certificate has been issued to the warehouse from the government regulatory body
- **wh_breakdown_l3m**: Number of times the warehouse faced a breakdown in the last 3 months. Like a strike by workers, a flood, or an electrical failure
- **govt_check_l3m**: Number of times government officers have visited the warehouse to check the quality and expiry of stored food in the last 3 months
- **product_wg_ton**: Product has been shipped in the last 3 months. Weight is in tons

Understand Business Problems

1)Need of the study/project

This project data consists of 25,000 warehouse observations (24 variables), with shipment volume (product_wg_ton) varying significantly, ranging from approximately 2,000 to 55,000 tons, with a median of approximately 22,000 tons.

In other words, in situations where demand imbalances across regions and warehouses are significant, empirically-based allocation of inventory could result in overstocking in some locations (increased disposal and storage costs) and out-of-stocks (loss of sales) in others. Therefore, demand forecasting and shipment volume decision-making (optimization) utilizing past sales and supply data are necessary.

2)Business opportunity

From a business perspective, simply reducing excess inventory can significantly reduce inventory holding costs. Inventory holding costs are often cited in the industry as *20-30% of inventory value. (*Reference 1)

Furthermore, from a logistics cost perspective, India's logistics costs are estimated at *7.97% of GDP (2023-24) at the national level, demonstrating that logistics efficiency is directly linked to corporate competitiveness. Therefore, creating a system to predict and distribute optimal shipment volumes for each warehouse offers opportunities to improve cost structures by reducing unnecessary movement and inventory, while simultaneously reducing stock-outs and boosting sales and customer loyalty. (*Reference 2)

3)Social opportunity

From a social perspective, excess inventory caused by supply-demand mismatch increases unsellable inventory, leading to increased product and packaging waste (especially plastic), and additional logistics such as inventory transfers and relocation. In Korea, plastic waste generation is expected to increase to *12.6 million tons by 2022, increasing the need for packaging reduction and a transition to a circular economy. (*Reference 3)

Furthermore, logistics and transportation themselves are directly linked to greenhouse gas emissions. Research indicates that freight transportation accounts for approximately *8% of global greenhouse gas emissions (approximately 11% when warehouses and ports are included). Therefore, data-driven decisions on optimal shipment volumes for each warehouse can simultaneously reduce inventory disposal and unnecessary transportation, thereby creating social value and lowering the environmental burden. (*Reference 4)

India also addresses the scale of plastic waste as a major issue in public institutions and policy documents, with data citing CPCB (Central Pollution Control Board) statistics indicating that India's plastic waste is estimated at 9.46 million tonnes per year. (*Reference 5)

Loading the Dataset and Data Overview

The dataset provided was loaded into a pandas dataframe for analysis.

Getting the first 5 and last 5 rows

The dataset has been loaded successfully. It has 25000 rows and 24 columns.

- First 5 rows of the dataset:

	Ware_house_ID	WH_Manager_ID	Location_type	WH_capacity_size	zone	WH_regional_zone	num_refill_req_l3m	transport_issue_l1y	Competitor_in_mkt	retail_shop_num	electric_supply	dist_from_hub	workers_num	wh_est_year	storage_issue_reported_l3m	temp_reg_mach	approved_wh_govt_certificate	wh_breakdown_l3m	govt_check_l3m	product_wg_ton
0	WH_100000	EID_50000	Urban	Small	West	Zone 6	3	1	2	4651	1	91	29.0	NaN	13	0	A	5	15	17115
1	WH_100001	EID_50001	Rural	Large	North	Zone 5	0	0	4	6217	1	210	31.0	NaN	4	0	A	3	17	5074
2	WH_100002	EID_50002	Rural	Mid	South	Zone 2	1	0	4	4306	0	161	37.0	NaN	17	0	A	6	22	23137
3	WH_100003	EID_50003	Rural	Mid	North	Zone 3	7	4	2	6000	0	103	21.0	NaN	17	1	A+	3	27	22115
4	WH_100004	EID_50004	Rural	Large	North	Zone 5	3	1	2	4740	1	112	25.0	2009.0	18	0	C	6	24	24071

Table 1: First 5 rows of the dataset

- Last 5 rows of the dataset:

	Ware_house_ID	WH_Manager_ID	Location_type	WH_capacity_size	zone	WH_regional_zone	num_refill_req_l3m	transport_issue_l1y	Competitor_in_mkt	retail_shop_num	...	electric_supply	dist_from_hub	workers_num	wh_est_year	storage_issue_reported_l3m	temp_reg_mach	approved_wh_govt_certificate	wh_breakdown_l3m	govt_check_l3m	product_wg_ton
24995	WH_124995	EID_74995	Rural	Small	North	Zone 1	3	0	4	5390	...	1	142	34.0	2005.0	22	1	A	2	30	32093
24996	WH_124996	EID_74996	Rural	Mid	West	Zone 2	6	0	4	4490	...	1	130	28.0	2012.0	10	0	B	4	18	12114
24997	WH_124997	EID_74997	Urban	Large	South	Zone 5	7	0	2	5403	...	1	147	NaN	NaN	23	0	B+	5	25	27080
24998	WH_124998	EID_74998	Rural	Small	North	Zone 1	1	0	2	10562	...	1	60	25.0	NaN	18	0	A	6	30	25093
24999	WH_124999	EID_74999	Rural	Mid	West	Zone 4	8	2	4	5664	...	1	239	39.0	2019.0	4	0	B+	2	11	5058

5 rows × 24 columns

Table 2: Last 5 rows of the dataset

Information about the dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25000 entries, 0 to 24999
Data columns (total 24 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   Ware_house_ID                25000 non-null  object
 1   WH_Manager_ID                25000 non-null  object
 2   Location_type                25000 non-null  object
 3   WH_capacity_size             25000 non-null  object
 4   zone                         25000 non-null  object
 5   WH_regional_zone             25000 non-null  object
 6   num_refill_req_l3m           25000 non-null  int64
 7   transport_issue_l1y          25000 non-null  int64
 8   Competitor_in_mkt            25000 non-null  int64
 9   retail_shop_num              25000 non-null  int64
 10  wh_owner_type                25000 non-null  object
 11  distributor_num              25000 non-null  int64
 12  flood_impacted               25000 non-null  int64
 13  flood_proof                  25000 non-null  int64
 14  electric_supply              25000 non-null  int64
 15  dist_from_hub                25000 non-null  int64
 16  workers_num                  24010 non-null  float64
 17  wh_est_year                  13119 non-null  float64
 18  storage_issue_reported_l3m   25000 non-null  int64
 19  temp_reg_mach                25000 non-null  int64
 20  approved_wh_govt_certificate 24092 non-null  object
 21  wh_breakdown_l3m             25000 non-null  int64
 22  govt_check_l3m               25000 non-null  int64
 23  product_wg_ton               25000 non-null  int64
dtypes: float64(2), int64(14), object(8)
memory usage: 4.6+ MB
```

Table 3: Information about the dataset

Checking for Duplicate values

There are no duplicate values in the dataset.

Data Checking

- I checked what values were in the categorical data and there were no special characters or typos

- Missing data analysis revealed a large number of missing data (11,881 observations approximately 47.5%) in wh_est_year, 990 missing observations (approximately 4.0%) in workers_num, and 908 missing observations (approved_wh_govt _certificate) out of a total of 25,000 observations. The remaining variables had no missing data.
  In particular, wh_est_year had a high percentage of missing data, which could lead to bias when simply imputed.

Statistical Description of the Dataset

A description of the columns of the dataset is given below:

	count	mean	std	min	25%	50%	75%	max
num_refill_req_l3m	25000.0	4.089040	2.606612	0.0	2.0	4.0	6.0	8.0
transport_issue_l1y	25000.0	0.773680	1.199449	0.0	0.0	0.0	1.0	5.0
Competitor_in_mkt	25000.0	3.104200	1.141663	0.0	2.0	3.0	4.0	12.0
retail_shop_num	25000.0	4985.711560	1052.825252	1821.0	4313.0	4859.0	5500.0	11008.0
distributor_num	25000.0	42.418120	16.064329	15.0	29.0	42.0	56.0	70.0
flood_impacted	25000.0	0.098160	0.297537	0.0	0.0	0.0	0.0	1.0
flood_proof	25000.0	0.054640	0.227281	0.0	0.0	0.0	0.0	1.0
electric_supply	25000.0	0.656880	0.474761	0.0	0.0	1.0	1.0	1.0
dist_from_hub	25000.0	163.537320	62.718609	55.0	109.0	164.0	218.0	271.0
workers_num	24010.0	28.944398	7.872534	10.0	24.0	28.0	33.0	98.0
wh_est_year	13119.0	2009.383185	7.528230	1996.0	2003.0	2009.0	2016.0	2023.0
storage_issue_reported_l3m	25000.0	17.130440	9.161108	0.0	10.0	18.0	24.0	39.0
temp_reg_mach	25000.0	0.303280	0.459684	0.0	0.0	0.0	1.0	1.0
wh_breakdown_l3m	25000.0	3.482040	1.690335	0.0	2.0	3.0	5.0	6.0
govt_check_l3m	25000.0	18.812280	8.632382	1.0	11.0	21.0	26.0	32.0
product_wg_ton	25000.0	22102.632920	11607.755077	2065.0	13059.0	22101.0	30103.0	55151.0

Table 4: Statistical Summary of the dataset

From the descriptive statistics, we can conclude that:

- While the average and median values for product _wg_ton are similar, the variability is significant , highlighting the need for differentiated policies based on regional and warehouse characteristics.

- Operational risk/ facility variables, such as flood _impacted (approximately 9.8%) and temp_reg_ mach (approximately 30.3%), are concentrated in specific warehouse groups, necessitating risk-based shipping and inventory policies.

- Demand channel scale variables, such as the number of retailers and distributors, exhibit high variance and a high-level range. This suggests the potential for increased shipment volume in the high - level range, necessitating focused management of high- demand locations.

Data Preprocessing

	0
Ware_house_ID	0
WH_Manager_ID	0
Location_type	0
WH_capacity_size	0
zone	0
WH_regional_zone	0
num_refill_req_l3m	0
transport_issue_l1y	0
Competitor_in_mkt	0
retail_shop_num	0
wh_owner_type	0
distributor_num	0
flood_impacted	0
flood_proof	0
electric_supply	0
dist_from_hub	0
workers_num	990
wh_est_year	11881
storage_issue_reported_l3m	0
temp_reg_mach	0
approved_wh_govt_certificate	908
wh_breakdown_l3m	0
govt_check_l3m	0
product_wg_ton	0

dtype: int64

Table 5:Missing value before data preprocessing

	feature	n_levels	F	p_value
5	approved_wh_govt_certificate	6	623.694905	0.000000e+00
9	temp_reg_mach	2	260.930561	2.126171e-58
0	Location_type	2	139.770772	3.639843e-32
1	WH_capacity_size	3	1.105525	3.310533e-01
3	WH_regional_zone	6	1.144966	3.339541e-01
2	zone	4	1.105561	3.453489e-01
4	wh_owner_type	2	0.843020	3.585434e-01
6	flood_impacted	2	0.132118	7.162490e-01
8	electric_supply	2	0.116611	7.327428e-01
7	flood_proof	2	0.004872	9.443548e-01

Table 6: ANOVA results for categorical data

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25000 entries, 0 to 24999
Data columns (total 24 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Ware_house_ID                 25000 non-null  object
 1   WH_Manager_ID                 25000 non-null  object
 2   Location_type                 25000 non-null  object
 3   WH_capacity_size              25000 non-null  object
 4   zone                          25000 non-null  object
 5   WH_regional_zone              25000 non-null  object
 6   num_refill_req_l3m            25000 non-null  int64
 7   transport_issue_l1y           25000 non-null  int64
 8   Competitor_in_mkt             25000 non-null  int64
 9   retail_shop_num               25000 non-null  int64
 10  wh_owner_type                 25000 non-null  object
 11  distributor_num               25000 non-null  int64
 12  flood_impacted                25000 non-null  int64
 13  flood_proof                   25000 non-null  int64
 14  electric_supply               25000 non-null  int64
 15  dist_from_hub                 25000 non-null  int64
 16  workers_num                   25000 non-null  int64
 17  storage_issue_reported_l3m    25000 non-null  int64
 18  temp_reg_mach                 25000 non-null  int64
 19  approved_wh_govt_certificate  25000 non-null  object
 20  wh_breakdown_l3m              25000 non-null  int64
 21  govt_check_l3m                25000 non-null  int64
 22  product_wg_ton                25000 non-null  int64
 23  warehouse_age                 25000 non-null  int64
dtypes: int64(16), object(8)
memory usage: 4.6+ MB
```

Table 7: Data information after data preprocessing

- Missing values in the government certification variable (approved_wh_govt_certificate) should be interpreted as "applied but not yet issued," rather than simply omitted. Therefore, the missing values were replaced with a separate category called "Applied_not_received" to preserve their meaning.

- The warehouse establishment year (wh_est_year) was converted to warehouse age (warehouse_age), which is more interpretable, rather than used directly in the model input (base year = maximum year in the data + 1). After conversion, wh_est_year was removed to reduce redundant information.

- Warehouse_age showed a strong positive correlation (r=0.829, p<0.001) with the target (product_wg_ton), making it an important variable for prediction. Missing values for this variable were supplemented with KNN Imputation, utilizing related numerical variables (distance, number of retailers, number of competitors, refill/issue indicators, etc.) instead of simple mean/median imputation, to minimize information loss.
- Categorical variables were tested for target separation using ANOVA. In particular, government certification, temperature control device (temp_reg_mach), and location (location_type) were found to significantly explain the difference in average shipment volume (very small p-value).
- The number of workers (workers_num) was replaced with the median to ensure distribution stability and then standardized to integers. Numeric variables that could be converted to real numbers during the KNN imputation process were rounded and standardized to integers to ensure data consistency and interpretability.
- To verify the validity of preprocessing and feature engineering, we calculated the target separation power for each variable. Several numeric variables, such as storage_issue_reported_l3m and warehouse_age, and government certification (approved_wh_govt_certificate), demonstrated high separation power and were retained as core variables.
- Binary variables (flood_impacted, flood_proof, electric_supply, temp_reg_mach) were normalized from 0/1 to Yes/No to ensure semantic consistency and encoding consistency.

[Numeric separation TOP 10]

	feature	skewness	spearman_rho	target_median_separation
10	storage_issue_reported_l3m	0.113345	0.989230	29934.0
14	warehouse_age	0.071230	0.790814	23034.5
12	wh_breakdown_l3m	-0.068026	0.339185	20022.0
1	transport_issue_l1y	1.610907	-0.169194	7470.0
11	temp_reg_mach	0.855960	0.096016	2930.5
9	workers_num	1.095082	-0.009607	992.0
0	num_refill_req_l3m	-0.075217	0.003105	80.5
4	distributor_num	0.015213	0.004900	58.5
8	dist_from_hub	-0.005999	-0.004944	58.5
13	govt_check_l3m	-0.363262	-0.008553	52.5

[Categorical separation]

	cat_feature	median_separation	top_cat	top_median	bottom_cat	bottom_median
5	approved_wh_govt_certificate	20998.0	A+	26088.0	Applied_not_received	5090.0
0	Location_type	3002.0	Urban	25074.0	Rural	22072.0
3	WH_regional_zone	1931.5	Zone 2	23061.0	Zone 1	21129.5
2	zone	1010.0	East	23081.0	South	22071.0
1	WH_capacity_size	62.5	Mid	22128.5	Small	22066.0
4	wh_owner_type	31.5	Rented	22118.0	Company Owned	22086.5

Table 8: Separation of all features from their targets

- Since the numerical and categorical variables showed the following separation for the target, basic EDA was conducted on all variables, and segment EDA and K-Means Clustering EDA were conducted on the categorical variables of additional EDA, such as " approved_wh_govt_certificate", "Location_type", "temp_reg_mach", "WH_regional_ zone", and "zone", and the numerical variables, such as "storage_issue_reported_l3m", "warehouse_age", "wh_breakdown_l3m", "transport_issue_l1y", and "workers_num".

Exploratory Data Analysis

Univariate Analysis

In this section, we will analyze the distribution of independent variables.

First, let us see how the target variable (`Product_wg_ton`) is distributed.

[Target Summary]

	value
count	25000.000000
mean	22102.632920
std	11607.755077
min	2065.000000
1%	3130.000000
5%	5128.000000
25%	13059.000000
50%	22101.000000
75%	30103.000000
95%	43113.000000
99%	51075.020000
max	55151.000000
skew	0.331631
mean_minus_median	1.632920

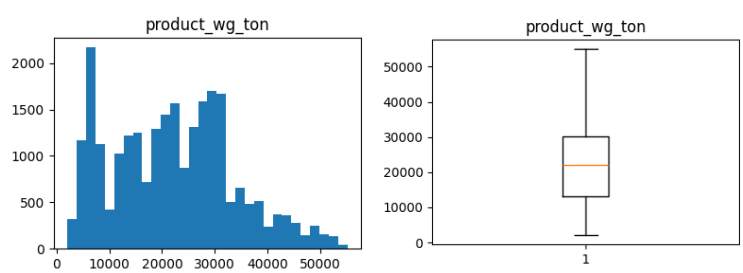


Fig 1: Observations on product_wg_ton

- Product_wg_ton fluctuates widely, ranging from approximately 2,000 to 55,000 tons, with a median of approximately 22, 000 tons. The distribution has a long right tail, meaning some warehouses handle extremely large volumes.

  Therefore, shipping policies should be differentiated by region and warehouse characteristics, with transportation and inventory stabilization policies prioritized for high- volume hubs.

Observations on Location_type

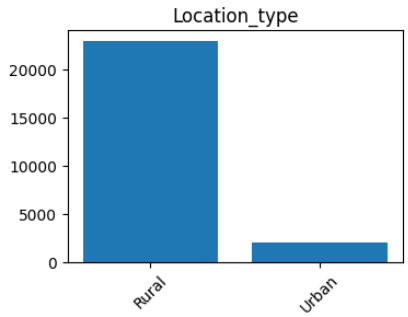


Fig 2: Observations on Location_type

- Location_type is overwhelmingly comprised of Rural warehouses, with Urban warehouses accounting for a relatively small proportion. In other words, the data represents a sample composition that more strongly reflects the operational characteristics of warehouses in rural locations.
- Logistics/inventory policies are likely to be standardized for rural-based warehouse operations. Urban warehouses, with their limited sample size, require further analysis and validation as separate operational scenarios.

Observations on zone

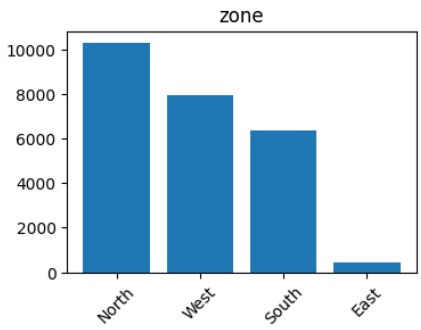


Fig 3: Observations on zone

- The North has the largest number of zones, followed by the West and South. The East has very few observations. This means that data is heavily concentrated in specific regions (North/West), and the East is a relatively rare sample.
- When interpreting regional demand/shipment patterns, the North/West region has a significant influence, while the East region's small sample size may lead to high estimation uncertainty. Therefore, from an operational perspective, it is reasonable to prioritize managing the North/West region as a "core region," while simultaneously securing additional data or implementing conservative decision-making (ex- safety stock , supplementary policies) for the East region.

Bivariate Analysis:

For bivariate analysis, we can analyze the contribution of variables in determining the `Product_wg_ton` which is the target variable.

Observations on storage_issue_reported_l3m and product_wg_ton

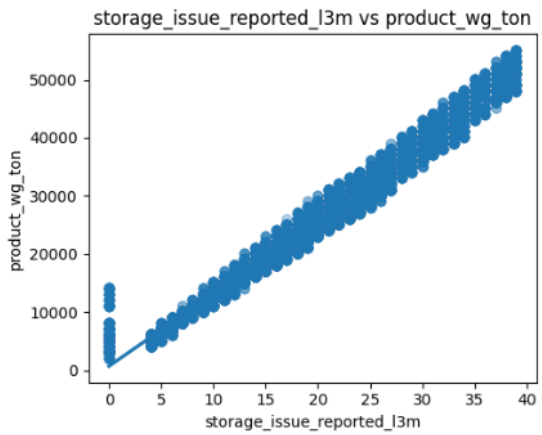


Fig 4: Observations on storage_issue_reported_l3m and product_wg_ton

- storage_issue_reported_l3m (the number of storage issues reported in the last 3 months) shows a very strong positive relationship with product_wg_ton (shipment volume), and a pattern is observed where the shipment volume increases almost linearly as the number of reports increases.
- This suggests that reports of storage issues may reflect the operational scale and activity of high-volume warehouses (high-shipment hubs) rather than simply the presence of a problem. Therefore, it would be reasonable to prioritize high-volume warehouses by focusing investments on strengthening the storage environment inspection cycle, temperature and humidity management, and preventative maintenance.

Observations on warehouse_age and product_wg_ton

Fig 5: Observations on warehouse_age and product_wg_ton

- A clear positive relationship is observed between warehouse_age and product_wg_ ton, and as age increases, the average shipment quantity tends to increase as well.
- This suggests that warehouses that have been in operation for a long time have likely grown into core hubs, accumulating regional logistics networks, clientele, and operational capabilities. Therefore, a reasonable strategy would be to prioritize older warehouses as "key hubs" and prioritize investments in facilities, personnel, and maintenance. For younger warehouses, the roles (buffer stock, regional distribution, etc.) would be gradually redesigned to reflect potential demand growth.

Observations on wh_breakdown_l3m and product_wg_ton

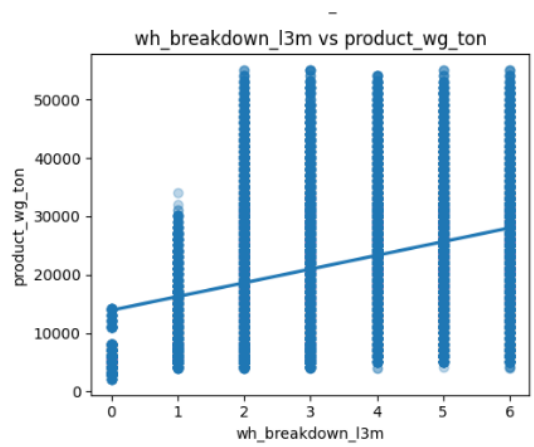


Fig 6: Observations on wh_breakdown_l3m and product_wg_ton

- A moderate positive relationship is observed between wh_breakdown_l3m (the number of warehouse outages/breakdowns in the last 3 months) and product_wg_ ton (shipments), and the average shipment volume tends to increase as the number of outages increases.
- This suggests that breakdowns and outages do not simply indicate inefficiency, but rather that core warehouses with high cargo volumes are experiencing increased operational burdens and increased exposure to incidents. Therefore, it makes sense to prioritize investments in stabilizing operating rates by targeting high-volume, high -outage warehouses.

Observations on transport_issue_l1y and product_wg_ton

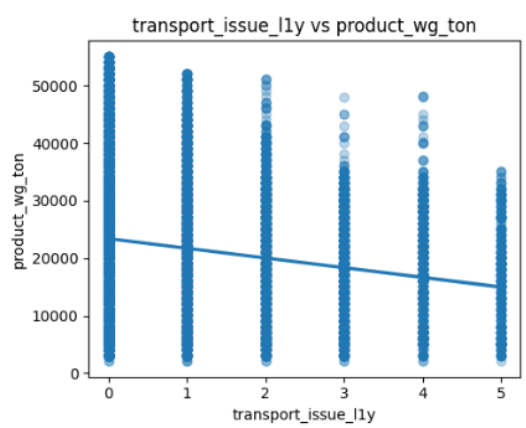


Fig 7: Observations on transport_issue_l1y and product_wg_ton

- A moderate negative correlation is observed between transport_issue_l1y (the number of transport issues in the past year) and product_wg_ton (shipments). The more transport issues there are, the lower the shipment volume tends to be.
- This suggests that as issues such as transport accidents, theft, and delays accumulate , supply stability may deteriorate, potentially limiting actual shipment execution. Therefore, warehouses with frequent transport issues should prioritize risk mitigation through carrier performance management, route/security enhancements, and securing alternative transport options.

Observations on approved_wh_govt_certificate and product_wg_ton

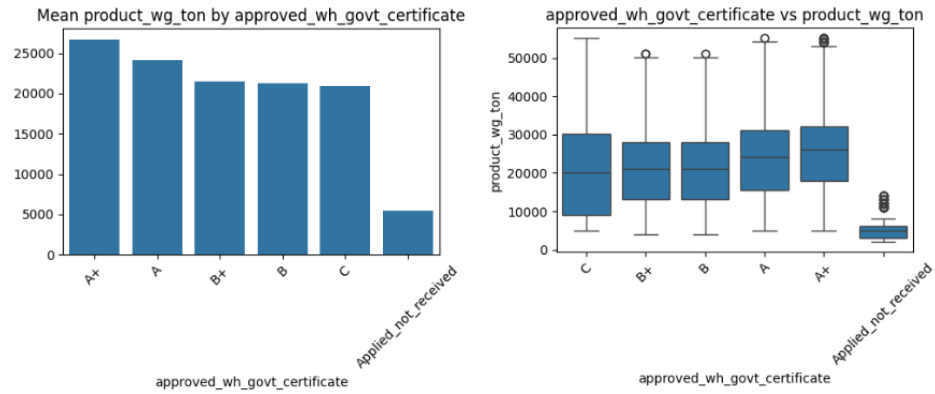


Fig 8: Observations on approved_wh_govt_certificate and product_wg_ton

- Comparing the average shipment volume (bar graph) by approved_wh_govt_ certificate grade, A+ and A grades show the highest levels, while B+/B/C are classified as lower. The boxplot also confirms a trend of higher median values as grades increase. In particular, the "Applied_not_received" group has significantly lower average and median values, and a narrower distribution, clearly distinguishing it from other grades.
- Therefore, government certification grades can be used as key indicators reflecting a warehouse's operational maturity and processing capabilities, and it is reasonable to prioritize support for obtaining certification (facility and process improvement) along with conservative allocation of inventory for 'Applied_not_received' warehouses.

Observations on temp_reg_mach and product_wg_ton

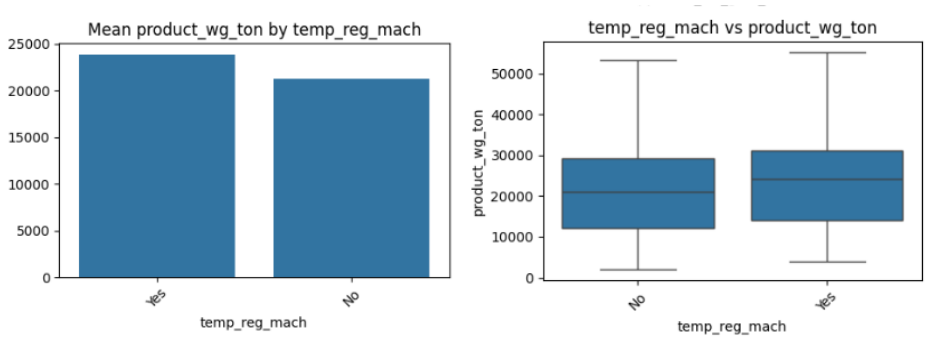


Fig 9: Observations on temp_reg_mach and product_wg_ton

- Looking at the average shipment volume by temp_reg_mach (whether or not temperature control equipment is available), warehouses with equipment (Yes) have higher average shipment volumes. The boxplot also shows that the median value for the "Yes" group is higher than that for the "No" group, suggesting that having temperature control equipment is associated with higher volumes (shipments). Therefore, it makes sense to prioritize (or expand) temperature control equipment in high-demand, high-shipment locations. For warehouses without temperature control equipment, it's wise to design a conservative product/volume allocation policy, taking quality risks into account.

Observations on Location_type and product_wg_ton

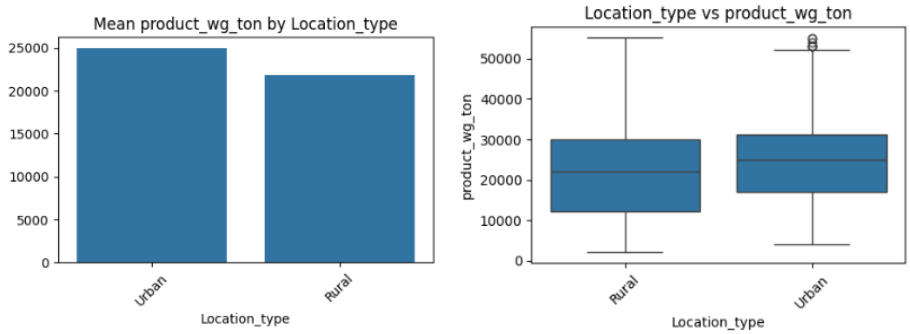


Fig 10: Observations on Location_type and product_wg_ton

- Comparing average shipment volumes by location type (Urban/Rural), urban warehouses exhibit higher average shipment volumes than rural warehouses. The boxplot also shows a higher median for the urban group, suggesting that urban warehouses likely handle greater demand (or greater distribution throughput). Therefore, it makes sense to prioritize urban areas as key demand hubs in inventory allocation strategies, while designing separate safety stock and replenishment cycles for rural areas, considering accessibility and lead times.
- The high proportion of rural warehouses, based on the number of observations, suggests a decentralized network. However, urban warehouses exhibit higher shipment volumes (mean/median), suggesting that urban warehouses have a relatively small number of locations handling larger volumes.

Multivariate Analysis:

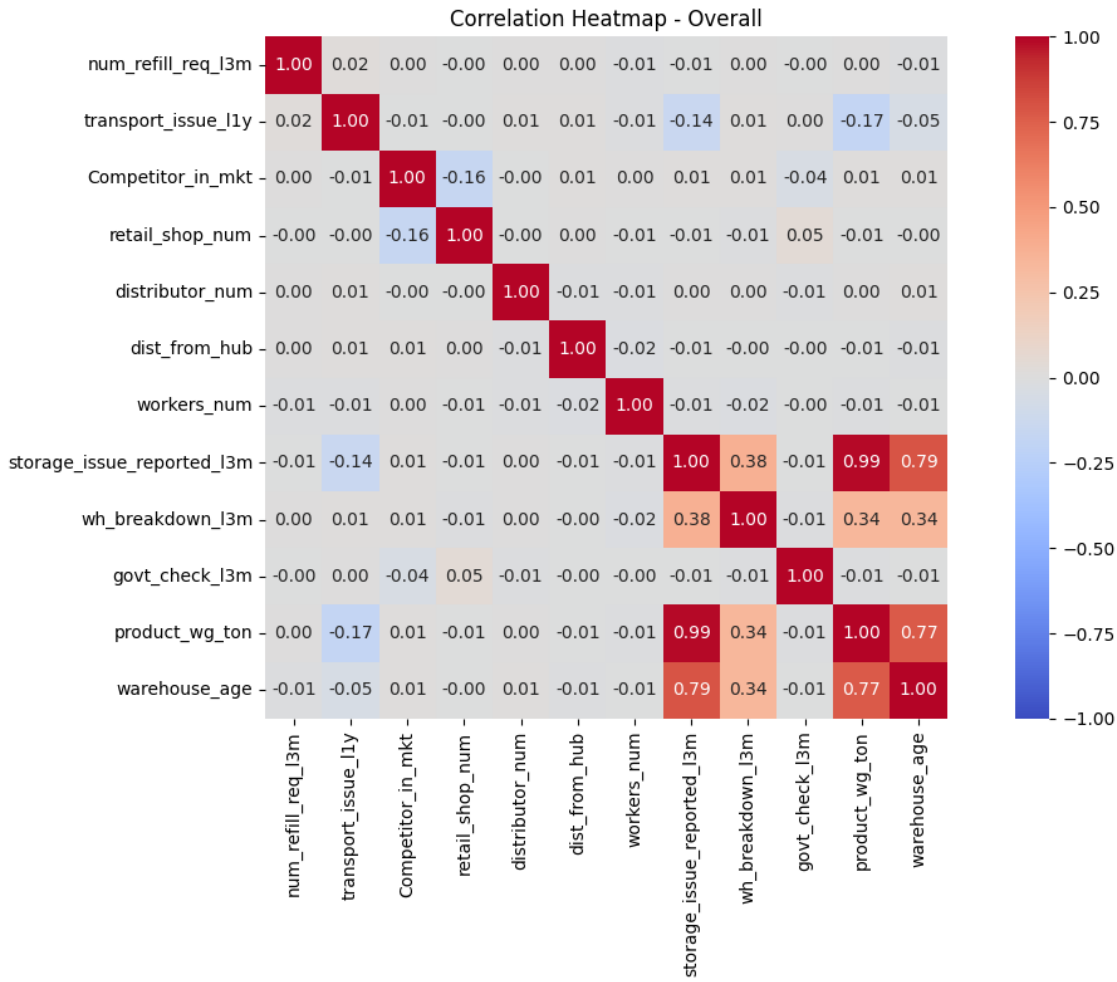


Fig 11: Correlation Analysis(Heatmap)

- A correlation analysis of numeric variables shows that shipment volume (product_wg_ton) has a very strong positive association with storage_issue_ reported_l3m (r=0.99) and a strong positive association with warehouse_ age (r=0.77). In addition, storage issues and warehouse age are also correlated (r=0.79), suggesting that older warehouses tend to report more storage issues. By contrast, transport_issue_l1y has a weak negative correlation with shipment volume (r=-0.17), implying that transport disruptions may be linked to lower executed shipment volumes. Given the high correlations among some predictors, subsequent linear modeling should include multicollinearity diagnostics (ex- VIF) and, if needed, feature selection.

Segmented EDA:

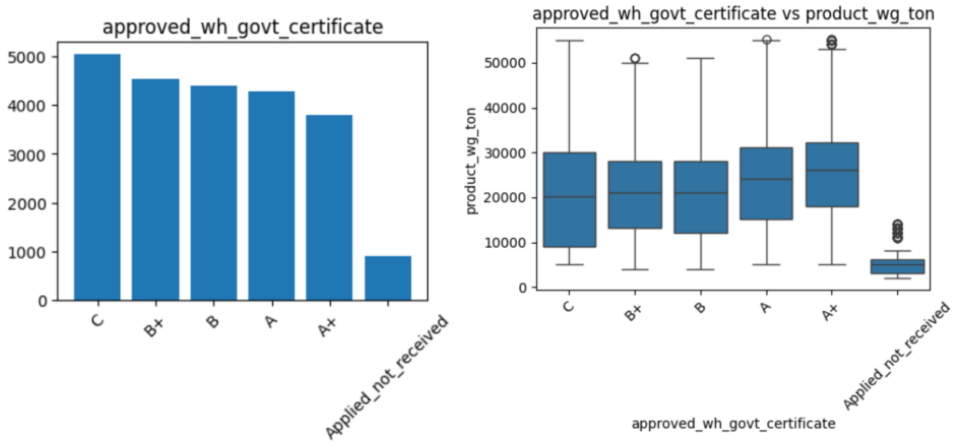


Fig 12: Segmented(Rural) EDA: approved_wh_govt_certificate

- In the Rural sector, the distribution of approved_wh_govt_certificates shows that most warehouses have government certification grades (A+/A /B+/B/C), with a small minority being "Applied_not_received." Therefore, in the Rural sector, most warehouses appear to have completed the certification process.

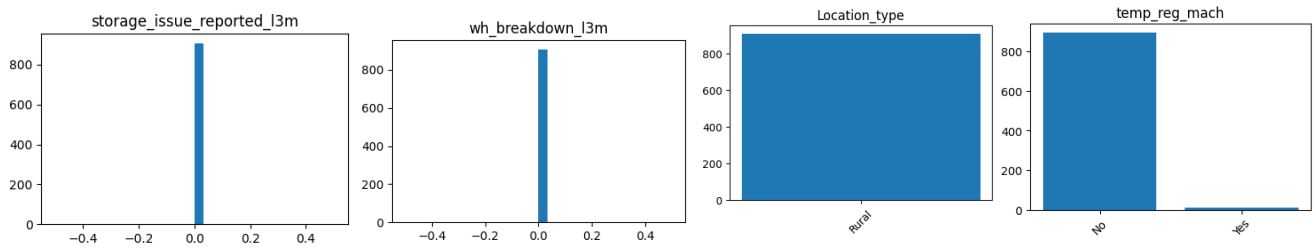


Fig 13: Segmented(Applied_not_received) EDA

- According to the EDA results for the government certification not issued (approved_wh_govt_certificate=Applied_not_received) segment, most temperature control equipment (temp_reg_mach) scores were "No," indicating that overall facility-based quality control capabilities may be low. Furthermore, the location (Location_type) is concentrated entirely in Rural areas, suggesting that the non-issuance issue may be a structural issue concentrated in a specific region/operating environment. Meanwhile, breakdowns (wh_breakdown_l3m) and storage issue reports (storage_issue_reported_l3m) in the past three months were observed to be almost all 0, suggesting that this may reflect an immature issue reporting system rather than actual stable operation. Therefore, for warehouses that have not been issued government certification, we propose the following as priorities: (1) establishing minimum facilities and operating standards, (2) reinforcing/replacing facilities when handling temperature-sensitive items, and (3) standardizing inspection and issue reporting processes.

Segmented EDA:

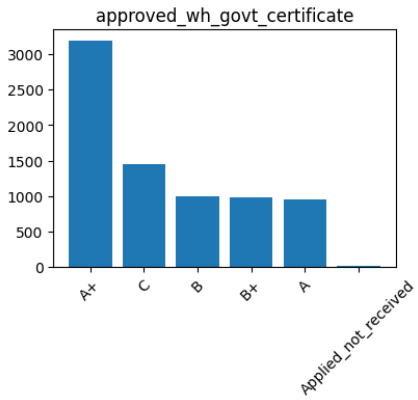


Fig 14: Segmented(temp_reg_mach=Yes) EDA: approved_wh_govt_certificate

- Looking only at warehouses with temperature control systems (temp_reg_mach=Yes), the government certification grades for A+ overwhelmingly dominate, suggesting that having facilities is associated with a high level of certification (operational standardization). However, mid- to lower-level grades, such as C, B, and B+, also exist in significant numbers, suggesting that certification is influenced not only by the presence of facilities but also by operational processes such as inspection, maintenance, and documentation.

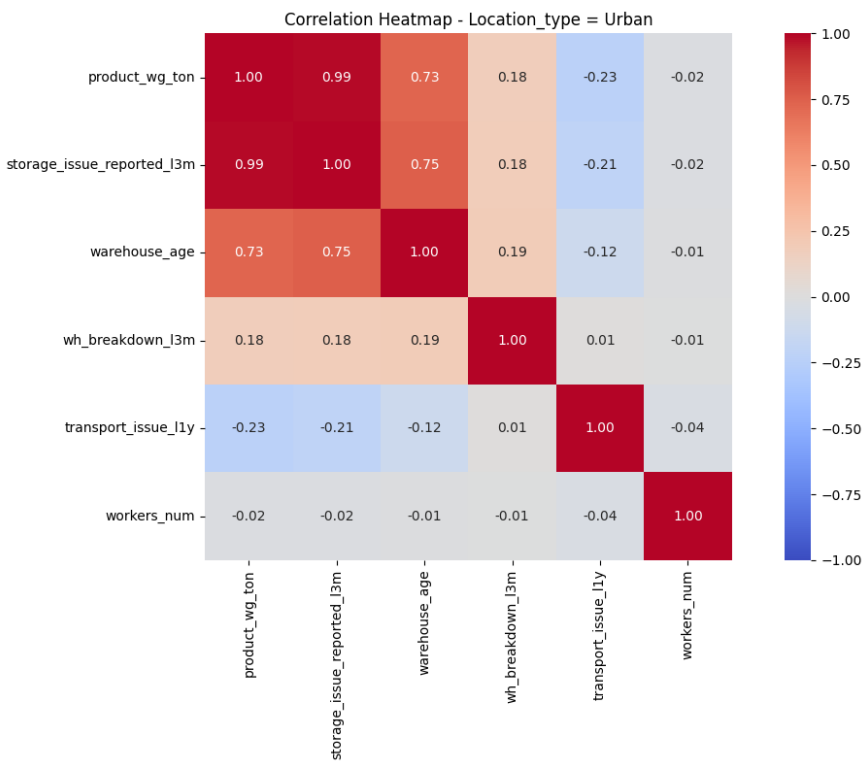


Fig 15: Segmented EDA: Correlation Analysis(Heatmap)

- In the urban segment, product_wg_ton still showed a very strong positive correlation (r=0.99) with storage_issue_reported_l3m and maintained a strong positive correlation (r=0.73) with warehouse_age. Conversely, transport_issue_l1y showed a stronger negative correlation (r=-0.23) compared to the overall score, suggesting that accumulated transport issues are more likely to impede material movement (supply execution) in urban areas. Furthermore, wh_breakdown_l3m showed a weaker correlation (r=0.18) compared to the overall score, suggesting that managing storage/obsolescence and transport risks is a more important m anagement priority than breakdown issues in urban areas.

K-Means Clustering EDA:

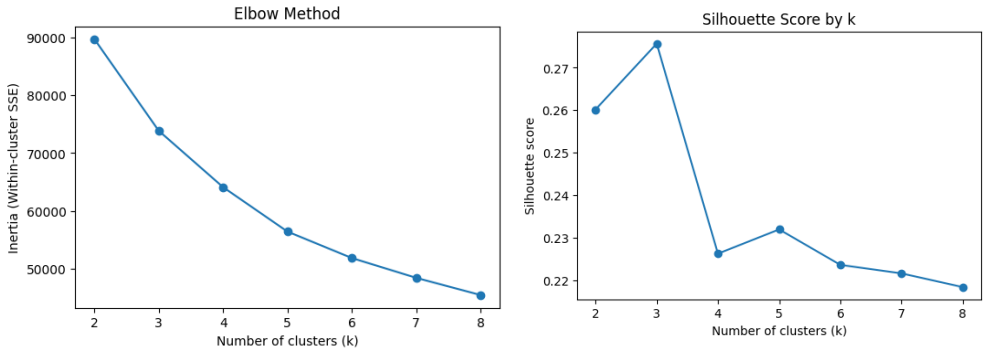


Fig 16: K-Means Clustering EDA_Elbow Method and Silhouette score

- In performed clustering, grouping similar hidden groups within the data. The k value for clustering was set to 3 using the elbow method and silhouette score, resulting in 3 groups.

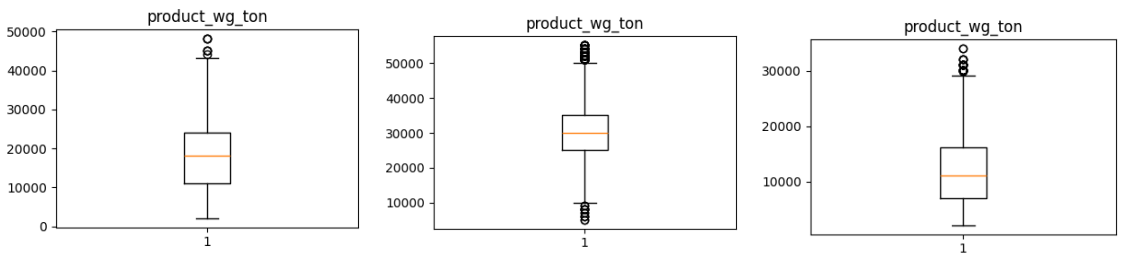


Fig 17: K-Means Clustering EDA: Each Cluster product_wg_ton

- The boxplot shows Cluster 0/Cluster 1/Cluster 2 from left to right. The warehouses are clearly divided into high (Cluster 1), medium (Cluster 0), and low (Cluster 2) operating groups based on cargo volume.

Cluster 0

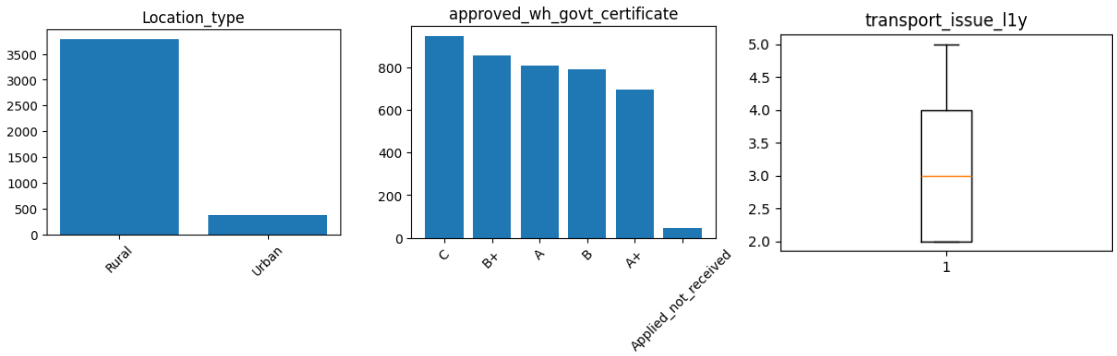


Fig 18: K-Means Clustering EDA: Cluster 0

- Cluster 0 has a high proportion of Rural, and certification is most frequently assigned to C. Furthermore, the average for transport issues (transport_issue_l1y) is relatively high, suggesting that improving the transport/shipping process is a priority.

Cluster 1

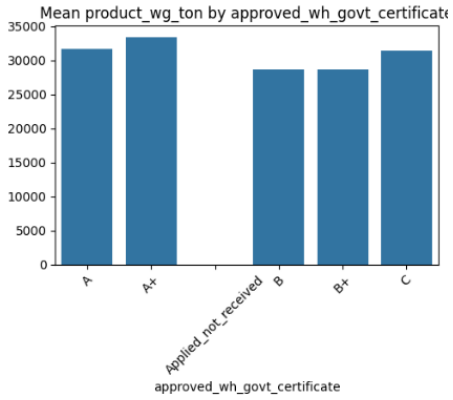


Fig 19: K-Means Clustering EDA: Cluster 1

- Cluster 1 has the highest volume (average: 30.7k) and serves as a key operational hub.Additionally, in this cluster, no Applied_not_received events were observed, indicating that most warehouses hold government certifications of grades A/A+/B/B+/C. This suggests a possible correlation between certification level and operational maturity.

Cluster 2

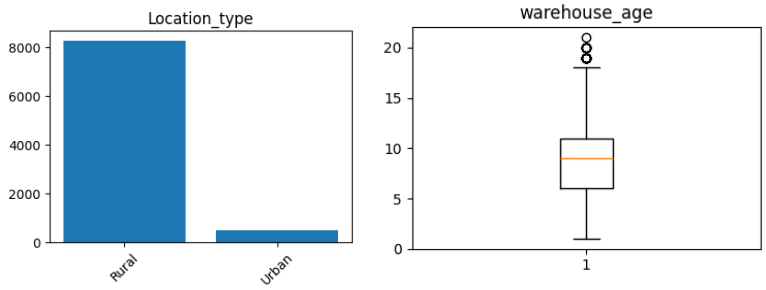


Fig 20: K-Means Clustering EDA: Cluster 2

- Cluster 2 has the lowest cargo volume (average: 12.1k), a high proportion of rural businesses, and relatively young warehouses (average: 8.5). An improvement strategy focused on onboarding, demand allocation, and establishing operational standards is more appropriate than facility investment.

Summary

- Cluster analysis results show that warehouses are divided into high (1), medium (0), and low (2) operating groups based on cargo volume, requiring differentiated management strategies. Operation strategies are categorized into Cluster 1 stabilization, Cluster 0 process improvement, and Cluster 2 onboarding and growth support.

Outlier Treatment

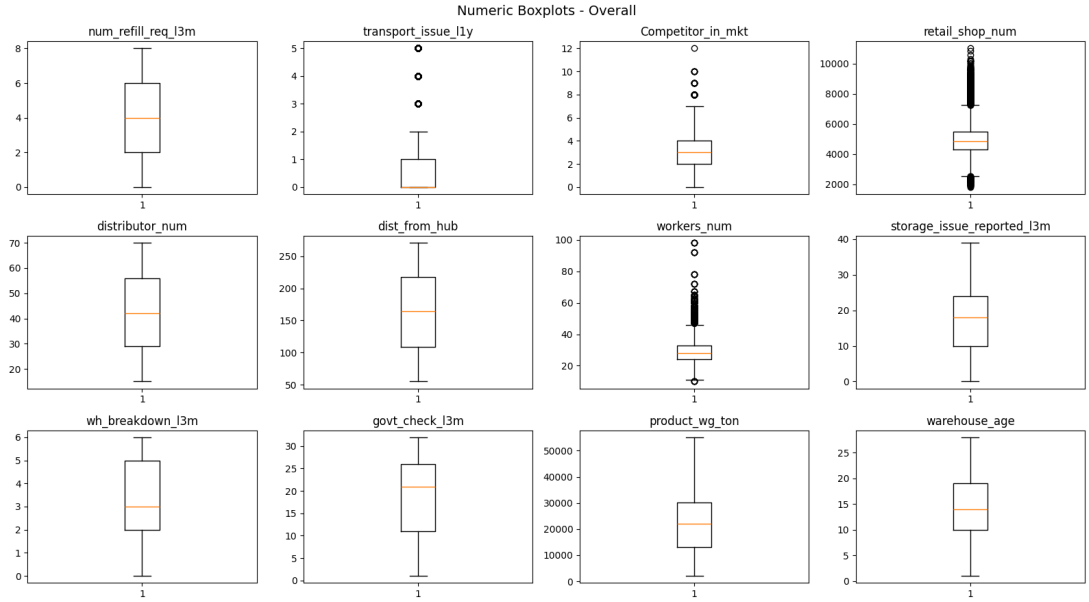


Fig 21: Before_Outlier Treatment

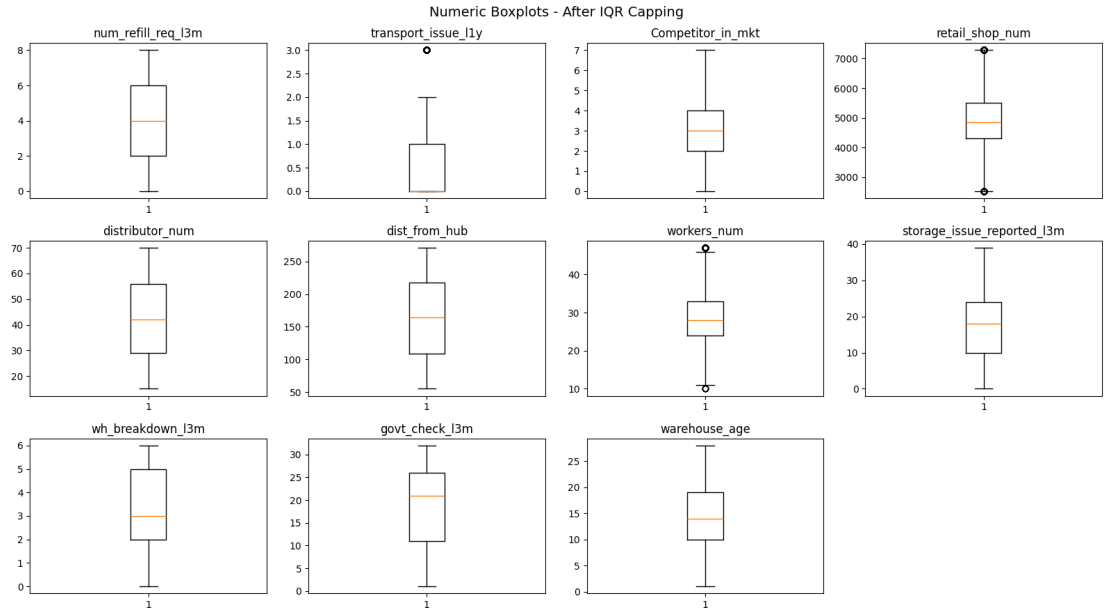


Fig 22: After_Outlier Treatment

- Boxplots revealed extreme values and long-tailed distributions in retail_shop_num, workers_num, Competitor_in_mkt, and transport_issue_l1y. This likely stems from differences in operational scale, with some warehouses serving large commercial districts and high-demand areas, making it difficult to dismiss this as a simple error . Therefore, without deleting outlier observations (to prevent information loss), we applied IQR-based capping to reduce their undue influence on model training. Values outside the lower and upper bounds calculated using the 1.5×IQR rule were replaced with the lower and upper bounds, respectively.

Model Building - Baseline Model

	model	n_features	train_RMSE	train_MAE	train_R2	train_AdjR2	test_RMSE	test_MAE	test_R2	test_AdjR2
0	Baseline OLS (all features, no VIF/p-iter)	32	1128.929014	870.282235	0.990557	0.990542	1126.205109	872.507405	0.990521	0.99046

Table 9: Baseline OLS result

- We first built a baseline linear regression (OLS) model as a reference point. To prevent data leakage, preprocessing was fitted only on the training set and then applied to the test set. The baseline achieved RMSE = 1126 tons, MAE = 872 tons , and R^2 = 0.990 on the test set. This baseline provides a benchmark to evaluate whether advanced models and hyperparameter tuning meaningfully improve prediction accuracy. RMSE was chosen as the primary metric because large shipment errors are more costly (stockouts or excess inventory) and RMSE penalizes large errors more heavily.

Baseline Improved

	iter	n_features	worst_p	worst_feature	val_RMSE	val_MAE	val_R2	val_AdjR2
0	0	32	0.992390	cat__WH_regional_zone_Zone 2	1128.439953	871.858653	0.990371	0.990309
1	1	31	0.982559	cat__WH_regional_zone_Zone 3	1128.439953	871.858653	0.990371	0.990311
2	2	30	0.913585	num__Competitor_in_mkt	1128.435935	871.855089	0.990371	0.990313
3	3	29	0.825848	num__retail_shop_num	1128.438077	871.850041	0.990371	0.990315
4	4	28	0.696654	num__govt_check_l3m	1128.390369	871.807292	0.990372	0.990318
5	5	27	0.698699	cat__WH_capacity_size_Mid	1128.377963	871.820367	0.990372	0.990320
6	6	26	0.508974	num__dist_from_hub	1128.315252	871.764935	0.990373	0.990323
7	7	25	0.356120	cat__flood_impacted_Yes	1128.273162	871.675872	0.990374	0.990326
8	8	24	0.312658	cat__WH_regional_zone_Zone 4	1128.259435	871.675845	0.990374	0.990328
9	9	23	0.409992	cat__WH_regional_zone_Zone 6	1128.285194	871.751877	0.990374	0.990329
10	10	22	0.341279	cat__WH_regional_zone_Zone 5	1128.296130	871.823998	0.990374	0.990331
11	11	21	0.309198	cat__wh_owner_type_Rented	1128.229062	871.838105	0.990375	0.990334
12	12	20	0.204530	cat__zone_South	1128.226390	872.013445	0.990375	0.990336
13	13	19	0.743448	cat__zone_North	1128.017917	871.881238	0.990378	0.990342
14	14	18	0.149515	cat__zone_West	1128.096896	871.902933	0.990377	0.990342
15	15	17	0.135101	cat__Location_type_Urban	1128.013522	871.836569	0.990378	0.990346
16	16	16	0.068260	cat__flood_proof_Yes	1127.840652	871.651945	0.990381	0.990350
17	17	15	0.070843	cat__WH_capacity_size_Small	1127.312830	871.382179	0.990390	0.990361
18	18	14	0.055377	num__workers_num	1127.038871	871.269864	0.990395	0.990368
19	19	13	0.082837	cat__electric_supply_Yes	1126.624612	870.790114	0.990402	0.990377
20	20	12	0.028737	num__distributor_num	1125.986629	870.224597	0.990413	0.990390

	model	robust_SE	n_features	test_RMSE	test_MAE	test_R2	test_AdjR2
0	OLS after backward elimination (p-value, val-m...	True	12	1126.181225	872.295289	0.990521	0.990498

```
[FINAL MODEL SUMMARY]
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.991
Model:                            OLS   Adj. R-squared:                  0.991
Method:                 Least Squares   F-statistic:                 1.436e+05
Date:                Sat, 10 Jan 2026   Prob (F-statistic):               0.00
Time:                        06:16:36   Log-Likelihood:            -1.6897e+05
No. Observations:               20000   AIC:                         3.380e+05
Df Residuals:                   19987   BIC:                         3.381e+05
Df Model:                          12
Covariance Type:                  HC3
==========================================================================================================
                                                   coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------------------------------
const                                           1038.8456     37.111     27.993      0.000     966.110    1111.582
num__num_refill_req_l3m                           15.0141      3.213      4.674      0.000       8.718      21.311
num__transport_issue_1ly                        -279.0916      6.528    -42.755      0.000    -291.886    -266.297
num__distributor_num                               1.2213      0.498      2.454      0.014       0.246       2.197
num__storage_issue_reported_l3m                 1311.0615      1.599    819.716      0.000    1307.927    1314.196
num__wh_breakdown_l3m                            -30.7999      4.908     -6.276      0.000     -40.419     -21.181
num__warehouse_age                               -51.2847      2.109    -24.317      0.000     -55.418     -47.151
cat__temp_reg_mach_Yes                           635.1268     19.511     32.552      0.000     596.886     673.367
cat__approved_wh_govt_certificate_A+             130.9953     28.510      4.595      0.000      75.118     186.873
cat__approved_wh_govt_certificate_Applied_not_received  4703.8827     96.981     48.503      0.000    4513.803    4893.962
cat__approved_wh_govt_certificate_B             -2023.4405     23.331    -86.728      0.000   -2069.168   -1977.713
cat__approved_wh_govt_certificate_B+            -2057.9388     23.193    -88.731      0.000   -2103.396   -2012.481
cat__approved_wh_govt_certificate_C              -255.8258     22.610    -11.315      0.000    -300.140    -211.512
==============================================================================
Omnibus:                     4065.684   Durbin-Watson:                   1.984
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            18399.916
Skew:                           0.928   Prob(JB):                         0.00
Kurtosis:                       7.317   Cond. No.                         382.
==============================================================================
```

Fig 23: Baseline Improved OLS

- Starting from the full-feature OLS baseline, we performed backward elimination using HC3 robust standard errors. To avoid selection bias, the model was fit on the training set, monitored on a validation set, and the test set was used only once for final evaluation. This reduced the model from 32 to 12 features while maintaining similar performance (test RMSE = 1126 tons, R^2 = 0.991), resulting in a more parsimonious and interpretable linear model.

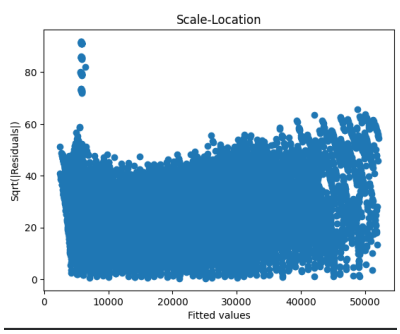
Comparison of Baseline Improved OLS

	RMSE	MAE	R2
mean	1127.602169	871.150315	0.990566
std	15.163722	8.790636	0.000265
min	1098.120561	859.026047	0.990118
max	1148.713599	886.160617	0.990973

Table 10: Comparison of improved OLS results

- To assess robustness, we repeated the train/test split across multiple random seeds using the same final 12-feature OLS model. Performance was stable: RMSE 1127.6 ± 15.2 tons, MAE 871.2 ± 8.8 tons, and R^2 0.9906 ± 0.0003, indicating the results are not driven by a single split.

Check OLS Assumption



	DurbinWatson	BP_LM_pvalue	BP_F_pvalue	White_LM_pvalue	White_F_pvalue
0	1.992244	0.0	0.0	0.0	0.0

Flags: 1279 / 20000 (Cook>0.000200 or |stud|>3 or lev>0.001300)

Fig 24: OLS Assumption Check Summary

- Residual plots show that errors spread out more when the predicted shipment is high, which means heteroskedasticity (unequal variance). Breusch–Pagan and White tests also confirm this (p = 0), while Durbin–Watson is near 2, so autocorrelation is not a concern. Therefore, we report OLS results with HC3 robust standard errors to make the conclusions reliable.

Model Building - Advanced Models

	Model	RMSE (metric of choice)	MAE	R2	MAPE(%)
0	GradientBoosting	872.542333	662.890510	0.994310	4.206715
1	Bagging	881.197354	655.573080	0.994196	4.088881
2	RandomForest	881.284335	656.046348	0.994195	4.090811
3	XGBoost	882.219467	665.053528	0.994183	4.256499
4	DecisionTree	995.450469	725.582130	0.992594	4.564358
5	AdaBoost	1705.171936	1383.272374	0.978269	10.035562

Table 11: Selection of tuning candidate models

- I trained and compared 6 advanced regression models (Decision Tree, Bagging, Random Forest, AdaBoost, Gradient Boosting, and XGBoost). I selected RMSE as the primary metric because large shipment errors (tons) should be penalized more. Among the models, Gradient Boosting achieved the best RMSE (= 872.5) with strong overall metrics (R^2 = 0.994), while Bagging/RandomForest/XGBoost performed similarly but slightly worse. AdaBoost underperformed (high RMSE/MAE), so it was not considered for final selection. B ased on this comparison, we chose the top-performing models (ex- Gradient Boosting , Random Forest, XGBoost) for hyperparameter tuning.

Model Performance Improvement using Hyperparameter Tuning

[RMSE improvement summary]

Model	Stage	Baseline	Tuned	RMSE_improvement(Baseline-Tuned)
XGBoost		884.046768	854.980373	29.066395
GradientBoosting		882.706324	860.204615	22.501709
RandomForest		888.558327	875.613970	12.944357

	Model	Stage	RMSE (metric of choice)	MAE	R2	MAPE(%)
0	GradientBoosting	Baseline	882.706324	672.297382	0.994177	4.303454
1	GradientBoosting	Tuned	860.204615	650.270886	0.994470	4.103631
2	RandomForest	Baseline	888.558327	662.300106	0.994099	4.162596
3	RandomForest	Tuned	875.613970	655.341912	0.994270	4.086147
4	XGBoost	Baseline	884.046768	664.627625	0.994159	4.257694
5	XGBoost	Tuned	854.980373	642.884094	0.994537	4.070367

Table 12: Comparison of RMSE before and after tuning

- I selected three high-performing models (Gradient Boosting, Random Forest, and XGBoost) for tuning because they capture non-linear patterns and typically benefit from hyperparameter optimization. To avoid data leakage, I tuned hyperparameters using 3-fold cross-validation on the training set only, and evaluated the tuned models once on the test set. Tuning improved RMSE for all three models. the largest gain was achieved by XGBoost (RMSE = from 884.05 to 854.98, improvement = 29.07), followed by Gradient Boosting and Random Forest. Therefore, XGBoost (tuned) is a strong candidate for the final model.

Model Performance Comparison and Final Model Selection

[Tuned model performance comparison on TEST]

	Model	Stage	RMSE (metric of choice)	MAE	R2	MAPE(%)
0	XGBoost	Tuned	854.980373	642.884094	0.994537	4.070367
1	GradientBoosting	Tuned	860.204615	650.270886	0.994470	4.103631
2	RandomForest	Tuned	875.613970	655.341912	0.994270	4.086147

Table 13: Final Model Selection: Tuning XGBoost

- After hyperparameter tuning, i compared the tuned candidates on the test set using multiple metrics, with RMSE as the metric of choice because large shipment errors (tons) are most costly (overstock or stockout).
Among the tuned models, XGBoost achieved the lowest RMSE (= 854.98) while maintaining strong overall accuracy (MAE = 642.88, R^2 = 0.9945, MAPE = 4.07 %). Therefore, i selected tuned XGBoost as the final model for deployment.

Actionable Insights & Recommendations:

- We will immediately implement a differential replenishment policy based on high-volume inventory. Volume varies significantly across warehouses (right-tailed distribution), and warehouse_age and storage_issue_reported_l3m in particular show a strong positive correlation with volume. Therefore, we will apply shorter replenishment cycles and increased replenishment caps (inventory caps) to high-volume hubs, while conservative allocations will be applied to low-volume warehouses.
- Government certification (approved_wh_govt_certificate) is used as a "policy trigger" for operational maturity. Warehouses rated A+/A are subject to increased target inventory levels, assuming high throughput. Conversely, warehouses rated Applied_not_received are subject to low average/median volumes and narrow distribution, making them more susceptible to "low demand and low maturity." Therefore, conservative shipping (focusing on safety stock) is initially implemented , along with support for certification (facility and process improvements).
- We introduce quality risk-based allocation rules based on the presence or absence of temperature-controlled equipment (temp_reg_mach). Warehouses with "Yes" equipment tend to handle higher throughput, so temperature-sensitive items are prioritized for placement in these warehouses. Warehouses with "No" equipment are prioritized for investment in equipment, with a conservative shipping volume, taking into account quality claims and scrap risks.
- Transport issues (transport_issue_l1y) are managed as "shipment execution risk ." As the number of transport issues tends to decrease actual shipment volume, the warehouse applies risk mitigation measures, including carrier KPIs (delays/ accidents/losses), securing alternative routes, strengthening security, and providing lead time buffers (negative impacts can be particularly significant in the urban segment).

- We standardize operational strategies for each cluster (3-track operation). Cluster analysis results show high (Cluster 1), medium (Cluster 0), and low (Cluster 2) volume groups. Therefore, we clearly allocate resources: Cluster 1 for stabilization, Cluster 0 for process improvement (reducing transportation and operational issues), and Cluster 2 for onboarding and growth support (demand distribution).
- The prediction model is directly linked to the "Shipping Decision." The final choice, tuned XGBoost, has the lowest RMSE (based on the test) (also showing excellent MAE, R^2, and MAPE) and is advantageous in reducing large errors. Therefore, it is operated as a decision rule in the form of the predicted shipment volume per warehouse ± safety factor (risk indicators: transportation/submergence/ temperature equipment/certification level) in the "Weekly/Monthly Shipping Plan."
- We expand operational KPIs to include "cost, quality, and carbon" to measure effectiveness. To reduce excess inventory, waste (especially packaging waste), and unnecessary transportation, we compare and quantify (1) disposal/inventory turnover, (2) quality claims, (3) urgent shipments/lead times, and (4) transportation issue rates before and after model implementation. We prioritize investment in high-performing segments (hub/urban/uncertified).

Actionable Insights & Recommendations-summary

- This analysis reveals that shipment volume (product_wg_ton) is strongly correlated with operational indicators such as warehouse_age (correlation r=0. 829, p<0.001) and storage_issue_reported_l3m, suggesting that differential allocation based on "warehouse maturity and operational issues" is most effective. In the Urban segment, the correlation between shipment volume and storage_issue_reported_l3m is particularly high (r=0.99), suggesting that proactive inspection and improvement of warehouses with accumulating issues will directly lead to shipment stabilization. Missing government certifications (approved_wh_govt_certificate) are defined as "Applied_not_received." Given that this segment has a high proportion of temperature-controlled equipment (temp_reg_mach=No) and tends to be concentrated in rural areas, establishing minimum operating standards and establishing an inspection and reporting system are recommended as priorities. Transport issues (transport_issue_l1y) act as an execution risk that limits cargo volume. Therefore, warehouses with high transport issues should implement risk mitigation measures such as managing carrier KPIs, securing alternative routes, and implementing lead time buffers. In terms of modeling, the tuned XGBoost performed the best on the test set with RMSE 854.98, MAE 642.88, and R^2 0.9945, suggesting that it be directly linked to decision-making in the form of “forecasted shipment volume ± safety factor (based on transportation/facility/certification/issues)” in the weekly/ monthly shipment plan. Lastly, the effect of policy introduction is quantified by comparing before and after with KPIs such as inventory turnover, disposal, out-of-stock, emergency shipment, and transportation accident rate, and is gradually expanded from segments with high performance.

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